



Mannich-like three-component synthesis of α -branched amines involving organozinc compounds: ReactIR monitoring and mechanistic aspects

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ABSTRACT

The infra-red spectroscopic *in situ* examination of the Mannich-like three-component reaction between organozinc reagents, amines and aldehydes is described, revealing that two distinct pathways are favored, depending on the nature of the starting amine. While reactions involving primary amines proceed via *in situ*-formed imines, reactions involving secondary amines rather imply a hemiaminal as the key intermediate species.

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1. Introduction

In the last decades, *in situ* spectroscopies have emerged as useful tools in chemistry by enabling reaction monitoring and providing mechanistic elucidations required for a deeper understanding of chemical processes, their optimization and development. Among a range of spectroscopic techniques available, e.g. NMR [1–4] or stopped-flow UV [5,6], the *in situ* mid-infra-red spectroscopy has emerged as a major technique in the area [7–12], providing access to accurate mechanistic interpretations under simple and useful (i.e., realistic) conditions. However, while physical chemistry processes provide many examples of investigated processes, the examination of modern organic reactions by FTIR techniques still remains underexplored. In this respect, multicomponent reactions would constitute especially challenging reaction systems for further *in situ* spectroscopic examinations as they represent complex chemical processes in organic chemistry, generally involving the formation of multiple transient species.

We recently described a three-component Mannich-like reaction between organozinc reagents, amines and aldehydes as a useful tool for the diversity-oriented synthesis of a range of nitrogen-containing structures like, e.g., diarylmethylamines, 1,2-

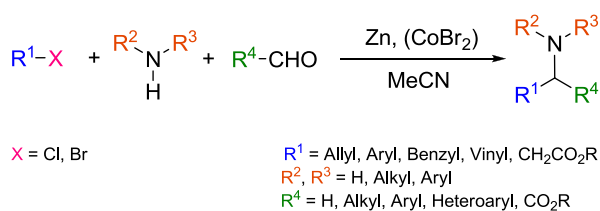
diarylethylamines, benzylamines, β -arylethylamines and α -amino esters [13,14]. While some excellent yields were obtained with this process, the reaction mechanism remains unclear, mainly due to the difficulty encountered in identifying the intermediate species. One of the main goals of our current research interests being directed towards the examination of mechanistic features for this three-component reaction, we therefore describe herein our first results obtained via ReactIR investigations. We assume that potential mechanistic pieces of information that would be gathered could open the way to the optimization of the reaction conditions and the possible development of stereoselective versions of the reaction.

2. Results and discussion

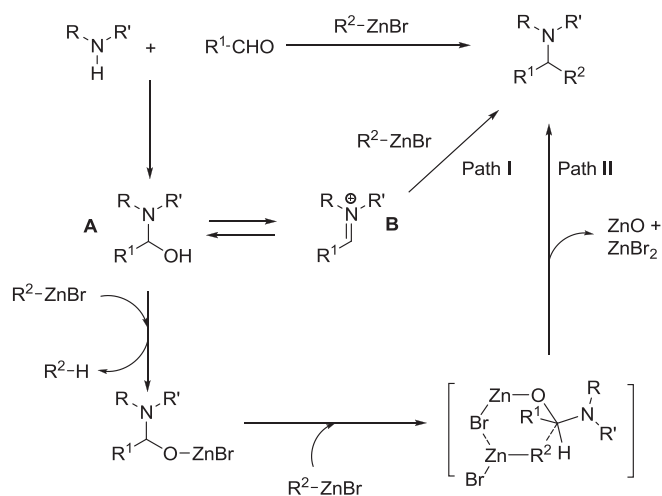
The investigated three-component reaction system is displayed in Scheme 1.

In this reaction, organozinc reagents are assumed to be formed *in situ* and to react with transient electrophilic species formed upon addition of the amine to the carbonyl compound. Therefore, the reaction can be considered as an “organometallic” variant of the Mannich reaction. However, while classical Mannich reactions are known to feature the formation of an iminium or an imine intermediate, depending on the nature of the starting amine, the system described herein could be more complex. Indeed, some published work indicates that Bruylants-type reactions of α -aminonitriles are

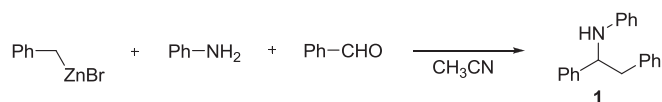
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Scheme 1. Multicomponent reaction investigated in this study.



Scheme 2. Proposed reaction pathways for the investigated multicomponent reaction.



Scheme 3. Model reaction for the examination of the mechanism when using primary amines.

possible when using organozinc reagents as nucleophiles [15–17]. Iminium ions could therefore constitute conceivable electrophiles in the process. On the other hand, a possible alternate mechanism has also been proposed by Fan and co-workers in a recent study on the multicomponent reaction of allyl- and benzylzinc halides with aldehydes and primary amines [18]. The authors of this study suggest that the imine formed upon reaction between the amine and the aldehyde is not the reactive electrophile in the process, but instead that the hemiaminal formed during the course of the reaction is deprotonated by the organozinc species and that the resulting zinc-alkoholate is subsequently engaged in a putative six-membered transition-state with a second equiv of the organozinc reagent to finally give rise to the formation of the three-component coupling product. Both reaction pathways are depicted in Scheme 2.

The preparation of the suggested reactive intermediates **A** and **B** and the examination of their reactions with organozinc reagents might constitute a possible, direct, experimental approach for the determination of the mechanism for this reaction. Unfortunately, while the formation of imines or iminium intermediates **B** (or synthetic equivalents thereof) might be considered, the preparation of acyclic hemiaminals such as **A** has not been described to date, probably due to an intrinsic instability. Consequently, the use of *in situ* FTIR spectroscopy is envisioned here as a practical experimental alternative to the preparation of the very unstable intermediates most probably involved in the reaction, assuming that these species could be detected and identified via specific vibrations, even at low concentrations. The requisite condition for a reliable study of the mechanism then calls for the unambiguous identification of specific absorption bands assignable to each species involved in the reaction (reagents, intermediates, products). In particular, these bands should not be mixed up with other absorption bands as they will serve as references for monitoring the evolution of the corresponding species in a qualitative (and semi-quantitative) manner during the reaction.

Two sets of substrates were selected as models in this study. First, the reaction of primary amines was investigated with aniline, benzaldehyde, and benzylzinc bromide as the organozinc species. For the monitoring of the reaction with secondary amines, piperidine was chosen instead, due the lack of reactivity of *N*-substituted aromatic amines under standard conditions. Furthermore, the

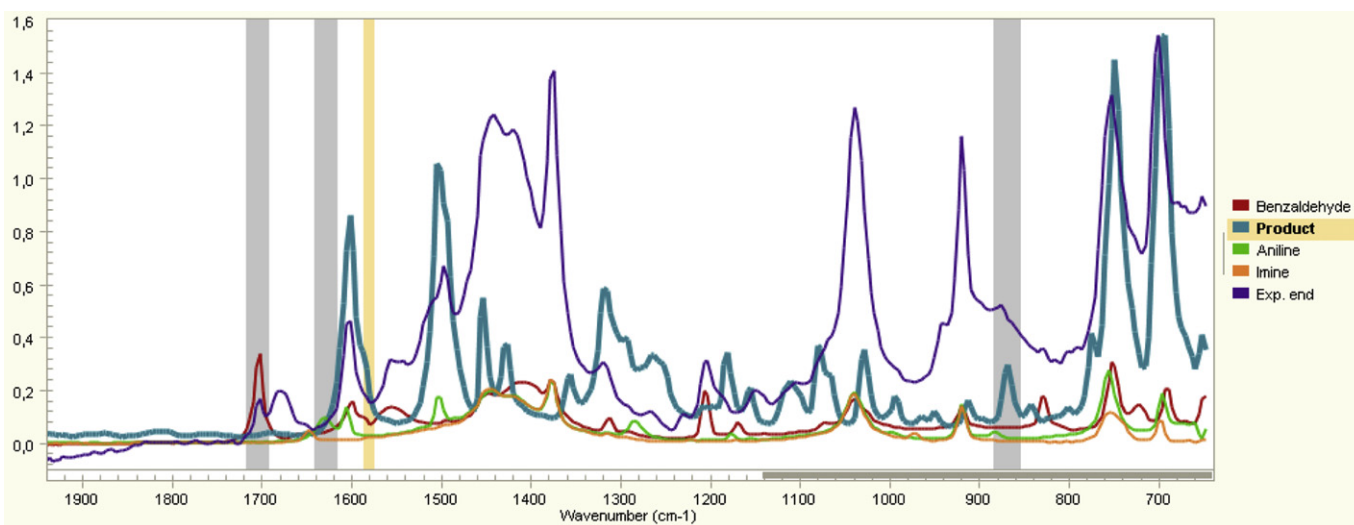
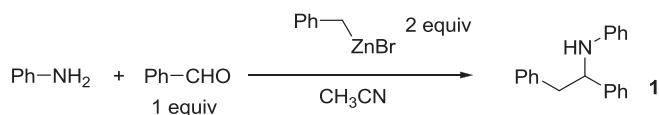


Fig. 1. Overlay of spectra obtained for two of the reagents (benzaldehyde (red) and aniline (green)), final product **1** (grey-blue), and putative imine intermediate PhCH=NPh (orange). The spectrum obtained at the end of the reaction (violet) when monitoring the process is added for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Scheme 4. Reaction scheme corresponding to the experimental protocol when using aniline.

benzylzinc reagent was replaced with an arylzinc reagent to both overcome increased spectral interferences [19] and further expand the scope of this work.

2.1. Monitoring of the reaction with primary amines

The mechanistic investigation was first realized with a primary amine, aniline, a preformed benzylzinc reagent and an aldehyde, benzaldehyde (Scheme 3). The reagents were chosen as “bare” as possible (i.e., non functionalized) to obtain simpler infra-red spectra and subsequently limit the number of absorption bands and the odds for spectral interferences.

Reagents and products of the reaction were individually characterized by IR, as well as a putative imine intermediate, *N*-benzylideneaniline ($\text{PhCH}=\text{NPh}$), that results from the *in situ* condensation of the aldehyde with the amine. Specific absorption bands were considered for each species, and selected as follows for the subsequent monitoring of the reaction: benzaldehyde: 1703 cm^{-1} , aniline: 1280 and 1630 cm^{-1} , imine: 1580 cm^{-1} , final product **1**: 870 cm^{-1} (Fig. 1).

The following experimental protocol was then applied: aniline and benzaldehyde (1 equiv each) were allowed to react for several minutes in acetonitrile, until preformed benzylzinc bromide [20] (2 equiv) was added dropwise to the solution (Scheme 4).

The monitoring, over time, of the absorbance associated to the specific bands mentioned above is depicted in Fig. 2.

At $t = 2$ min, aniline was introduced in the reaction medium (acetonitrile), followed by benzaldehyde at $t = 7$ min. Until $t = 37$ min, these species were allowed to react without further addition of reagents. During this time period, the formation of intermediates from aniline and benzaldehyde appears to be extremely limited, as evidenced by the absence of noticeable

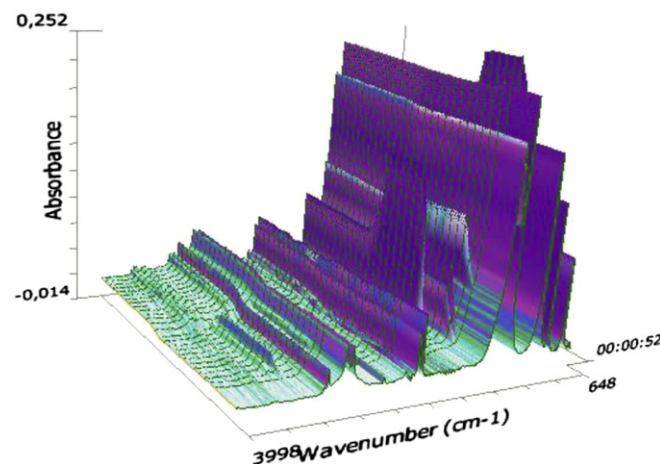


Fig. 3. Three-dimensional plot of the IR monitoring ($3998\text{--}648\text{ cm}^{-1}$ area) corresponding to the experimental reaction conditions summarized in Fig. 2.

modifications for the absorption of bands corresponding to benzaldehyde and aniline, respectively. Although both hemiaminal **A**₁ and imine **B**₁ could be plausible intermediates in the reaction, close inspection of the accumulated spectra only revealed the formation of imine **B**₁, no presence of OH bands attributable to the formation of hemiaminal **A**₁ being detected, as evidenced on the three-dimensional plot of the IR spectra monitoring (Fig. 3).

At $t = 37$ min, the organozinc reagent was introduced dropwise over a 10 min time period. As soon as the first drops of the solution were introduced in the reaction medium, the absorbance due to the imine and the coupling product **1** increased (mid-section in Fig. 2) whereas the intensity of the $\text{C}=\text{O}$ band rapidly decreased, associated to the consumption of benzaldehyde. It is worth noting that when aniline and benzaldehyde had first been introduced in the solvent (see previous paragraph), the reaction equilibrium between these reagents and the imine had only moderately shifted towards the imine formation, while the present introduction of the organozinc solution had the effect to significantly increase the imine formation, either by Lewis acid activation of the carbonyl function through the addition of zinc salts and/or by acid–base catalysis.

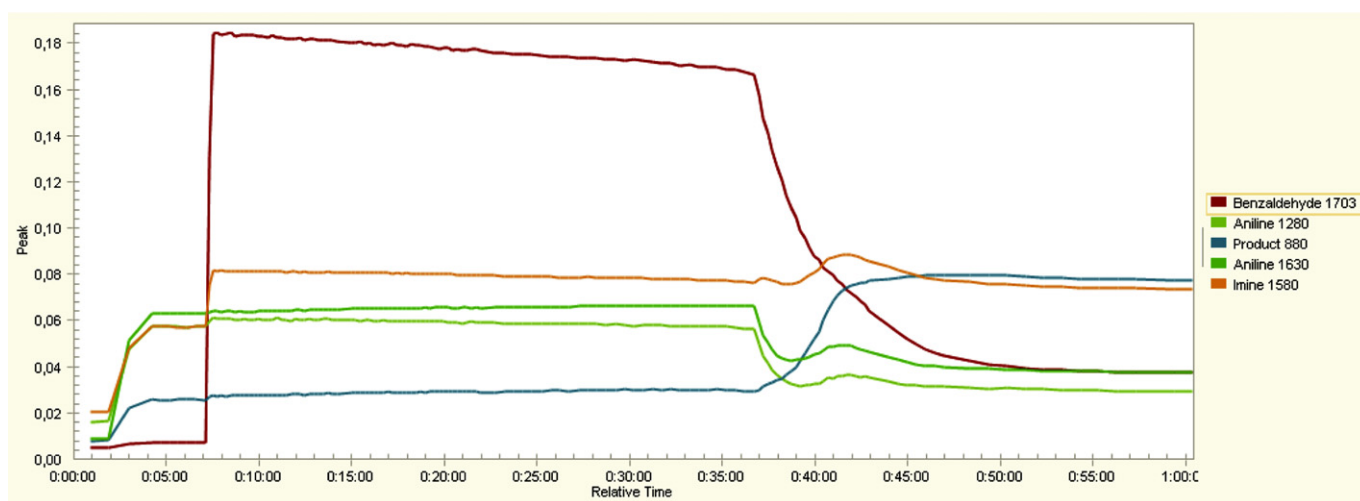


Fig. 2. Monitoring of the model three-component reaction when using aniline as a model primary amine; subsequent addition to acetonitrile used as a solvent of: (1) aniline (light-green (1280 cm^{-1} band) and green (1630 cm^{-1} band) curves), (2) benzaldehyde (red curve (1703 cm^{-1} band)), and (3) benzylzinc bromide; imine appearance is monitored at 1580 cm^{-1} (orange curve); product **1** appearance is monitored at 880 cm^{-1} (grey–blue curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

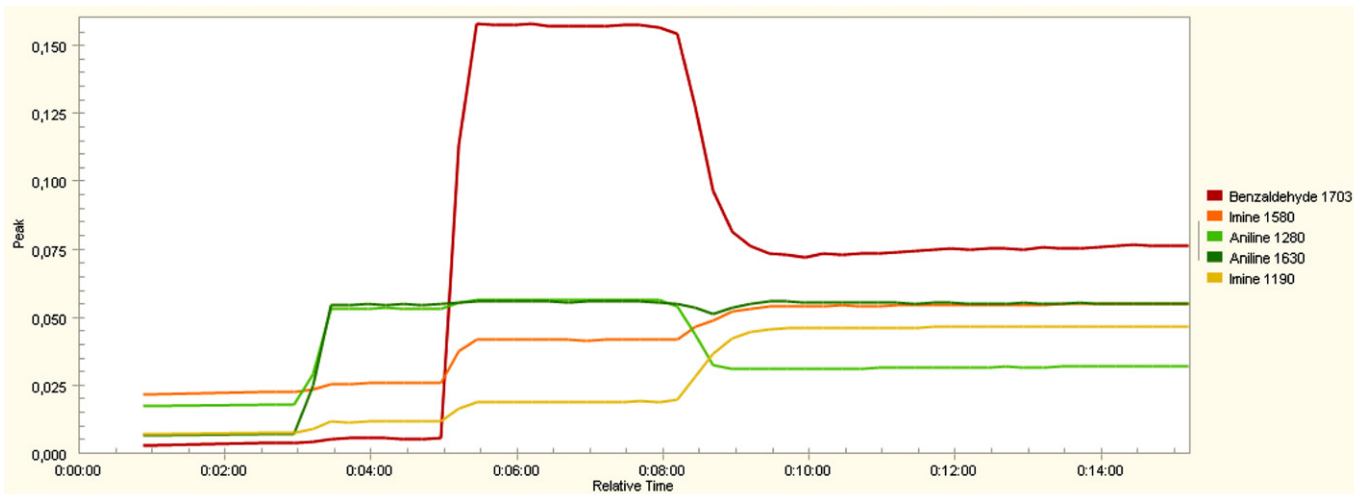
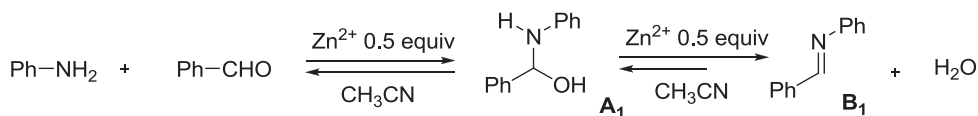


Fig. 4. Effect of zinc salts on the formation of imine **B₁** (light-orange (1190 cm^{-1} band) and orange (1580 cm^{-1} band) curves); subsequent addition to acetonitrile used as a solvent of: (1) aniline (light-green (1280 cm^{-1} band) and green (1630 cm^{-1} band) curves), (2) benzaldehyde (red curve (1703 cm^{-1} band)), and (3) zinc bromide. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Scheme 5. Formation of imine **B₁**.

In order to determine whether zinc salts contained in the solution were responsible for the rate enhancement observed in the imine formation, 0.5 equiv of zinc bromide were added to an acetonitrile solution containing benzaldehyde and aniline. The monitoring of this reaction is depicted in Fig. 4.

At $t = 3$ min, aniline was introduced in the reaction medium (acetonitrile), followed by benzaldehyde at $t = 5$ min. A slight increase in the imine **B₁** absorbance occurred, indicating that the imine is formed even in the absence of Lewis acids, albeit at low conversion. When 0.5 equiv of zinc bromide were added to the mixture at $t = 8.5$ min, a quite more significant

formation of the imine could be observed, along with a consistent decrease of benzaldehyde and aniline concentrations, demonstrating the rate-accelerating effect and equilibrium shift induced by zinc salts in the imine formation, as depicted in Scheme 5.

We then proceeded further with these reaction FTIR monitoring in order to verify that a mixture containing both reagents and zinc salts constitute a suitable reaction medium for the reaction of subsequently-added benzylzinc reagent. Thus, to the above solution was added preformed benzylzinc bromide. The corresponding monitoring curves are presented in Fig. 5.

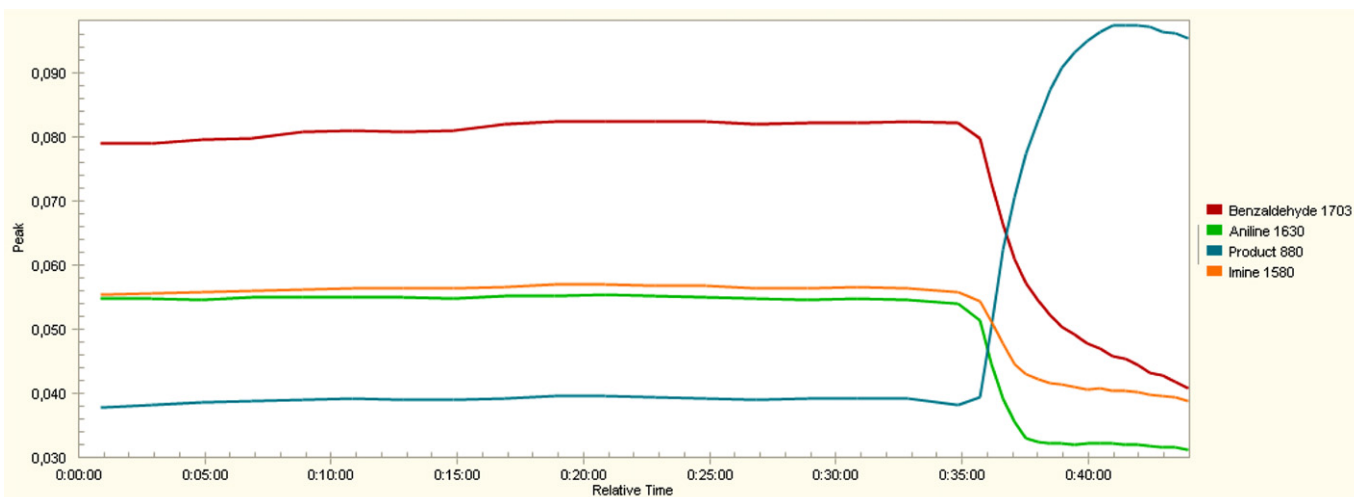
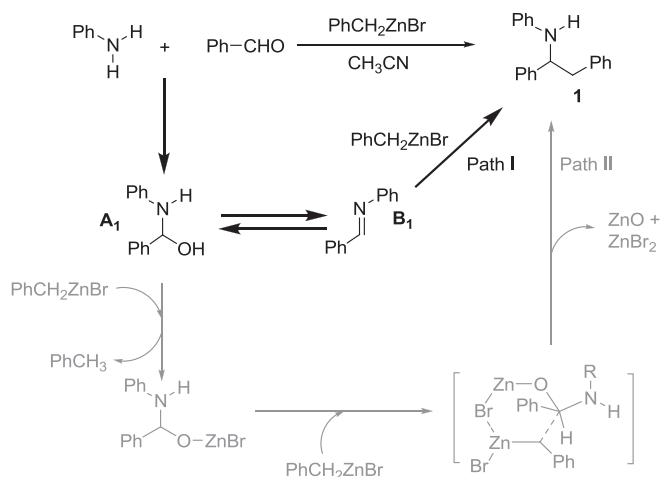
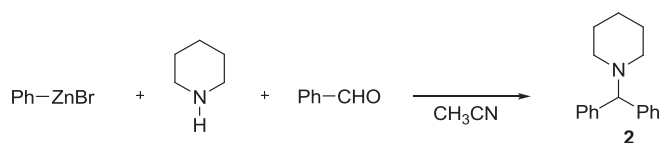


Fig. 5. Effect of the addition of benzylzinc bromide on an acetonitrile solution containing aniline (1 equiv, green curve (1630 cm^{-1} band)), benzaldehyde (1 equiv, red curve (1703 cm^{-1} band)) and zinc bromide (0.5 equiv); product **1** appearance is monitored at 880 cm^{-1} (grey-blue curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Scheme 6. Putative reaction pathway when primary amines are used.



Scheme 7. Model reaction for the examination of the mechanism when using secondary amines.

At $t = 35$ min, the preformed organozinc reagent was added dropwise over a 5 min time period. The intensity of the band assigned to coupling product **1** raises quickly as soon as the addition starts and then reaches a maximum, concomitantly to a parallel decrease of aniline, benzaldehyde and imine absorbances, thus confirming that product **1** is formed by consumption of imine **B₁** via equilibrium shifting.

In summary, this first part of our study indicates that an imine is likely the reactive electrophilic intermediate in the multicomponent reaction between a primary amine, an aromatic aldehyde and a

benzylzinc reagent. In addition, we could show that imine **B₁** formation is favored by the addition of an organozinc solution containing zinc salts, which most probably act as Lewis acids, thus providing a rate enhancement in the reaction leading to coupling product **1**. This finding is corroborated by some of our previous work that had indicated that, under similar conditions, aromatic organozinc reagents do not react with *in situ*-generated imines, since they probably do not exhibit sufficient reactivity [21]. Thus, path **I** in Scheme 6 may be the most probable reaction pathway when primary amines are employed in this three-component reaction.

2.2. Monitoring of the reaction with a model secondary amine

An analogous mechanistic study was then carried out, starting from secondary amines. For consistency, we had initially intended to use a *N*-substituted secondary aniline, yet a different amine structure had finally to be used, due to both a lack of reactivity exhibited by *N*-substituted secondary anilines, and the observation of more spectral interferences than in the previous study. In order to circumvent these issues and obtain reliable results, the mechanistic investigation was thus realized with piperidine, benzaldehyde and phenylzinc bromide (Scheme 7).

Reagents and product of the reaction were individually characterized by IR, as well as a commercial iminium salt, Eschenmoser's salt (Fig. 10). Specific absorption bands were identified for each species, and selected as follows for the subsequent monitoring of the reaction: benzaldehyde: 1703 cm^{-1} and piperidine: 780 cm^{-1} (Fig. 6). Unfortunately, it was not possible to identify specific absorption bands for product **2** whose formation was thus monitored by GC analysis.

Owing to the influence of the zinc salt as noticed in the first part of this study, we first attempted to determine whether the corresponding salts would also promote the formation of reaction intermediates when benzaldehyde is mixed with piperidine. The IR monitoring of the corresponding experiment is presented in Fig. 7.

At $t = 4$ min, piperidine was introduced in the reaction medium (acetonitrile). The ensuing addition of benzaldehyde at $t = 7.5$ min resulted in the appearance of three weak bands at 1274 , 1084 and 1006 cm^{-1} (Fig. 8) as well as of a large band at 3540 cm^{-1} . When zinc bromide (0.5 equiv) was added to the solution at $t = 12.5$ min,

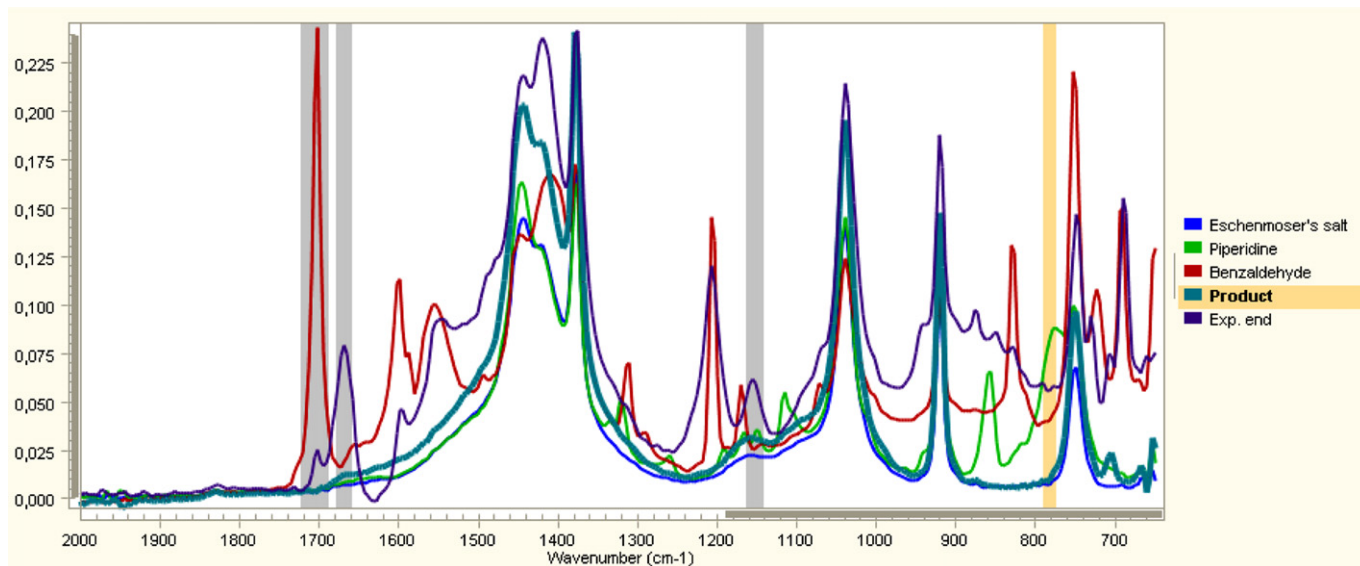


Fig. 6. Overlay of spectra obtained for two of the reagents (benzaldehyde (red) and piperidine (green)), final product **1** (grey-blue), and model iminium salt $\text{CH}_2=\text{N}^+\text{Me}_2$ (blue). The spectrum obtained at the end of the reaction when monitoring the process is added (violet curve) for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

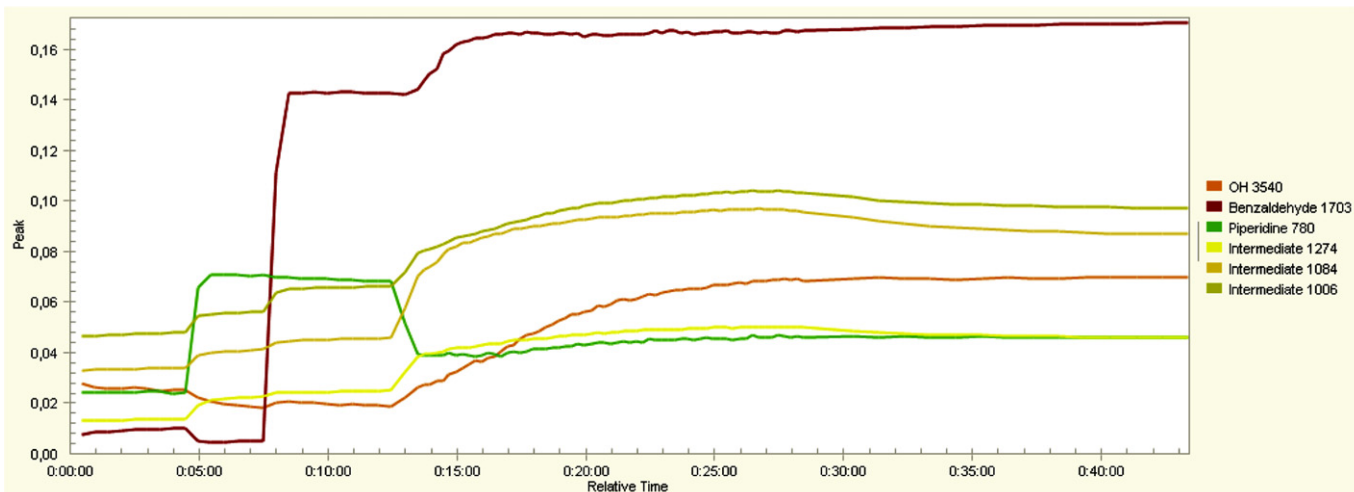


Fig. 7. Effect of zinc salts on the reaction between piperidine (green curve (780 cm^{-1} band)) and benzaldehyde (red curve (1703 cm^{-1} band)) in acetonitrile; the origin for extra curves (yellow (1274 cm^{-1} band), light-orange (1084 cm^{-1} band), orange (3540 cm^{-1} band) and light brown (1006 cm^{-1} band)) is discussed in the text. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the intensity of these bands further increased as the result of a possible Lewis-acid type activation of the carbonyl double bond and the resulting improved formation of a novel intermediate.

An examination of possible fits for the wavenumbers associated to the above bands suggests that a hemiaminal is likely formed upon mixing of piperidine and benzaldehyde. Indeed, hemiaminal **A**₁ would be characterized by typical vibrations for tertiary amines (1000–1200 cm^{-1}) and secondary alcohols (1050–1170 cm^{-1} and 1250–1350 cm^{-1}), in addition to the typical large band at 3400–3600 cm^{-1} that corresponds to the OH stretching of alcohols. It can be noted that the appearance of this latter band is clear, as attested by the three-dimensional plot corresponding to the full-wavelength IR monitoring of the reaction (Fig. 9).

Consequently, at first glance, it could be considered that zinc salts promote the formation of a hemiaminal intermediate from benzaldehyde and piperidine (Scheme 8).

Although the formation of hemiaminal **A**₂ appears consistent with the collected spectroscopic data, the *in situ* generation of an

iminium intermediate **B**₂PhCH=N(CH₂)₅⁺ via a further dehydroxylation of **A**₂ should also be considered. As typical stretching and bending vibrations have not been described for such species in the spectroscopic region of interest (i.e., 950–1300 cm^{-1}), we underwent the examination of the infra-red spectrum of a commercial iminium salt (Eschenmoser's salt) in acetonitrile to determine whether the observed vibrations could be attributed to iminium **B**₂ (Fig. 10).

An examination of the corresponding spectrum (Fig. 10) indicates that no significant bands are present in the three areas of interest. Thus, it can be concluded that experimental data strongly support the idea that zinc salts mainly promote the formation of hemiaminal **A**₂ rather than of iminium salt **B**₂. Consequently, it is rather difficult to consider the additional presence of an iminium species in the reaction medium, even in slight amounts.

The addition of an organozinc reagent on the above intermediate was then evaluated according to the following experiment: piperidine and benzaldehyde (1 equiv) were allowed to react for

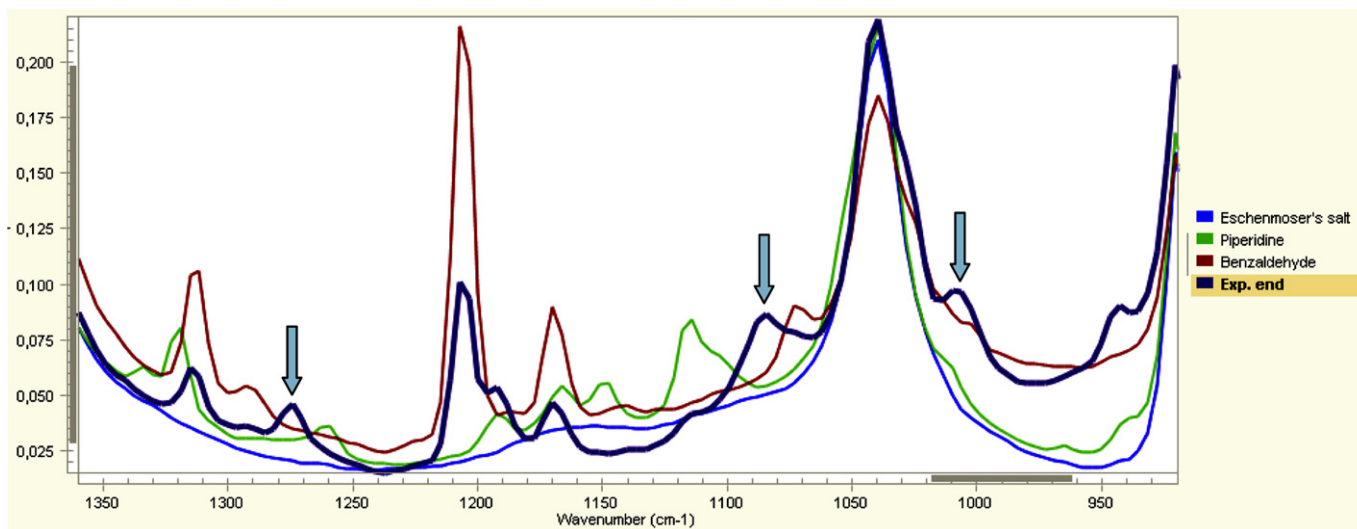


Fig. 8. Focus on the three new bands observed at 1274, 1084 and 1006 cm^{-1} (violet spectrum corresponding to the experiment's end) when mixing piperidine (green) and benzaldehyde (red) in acetonitrile; the spectrum for model iminium salts $\text{H}_2\text{C}=\text{N}(\text{CH}_3)_2$ (blue) is added for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

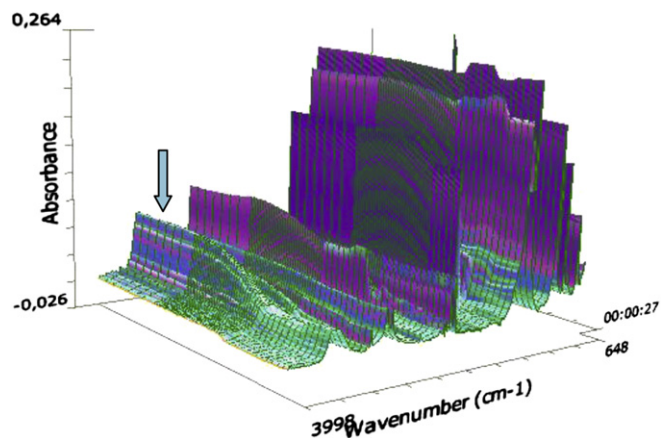
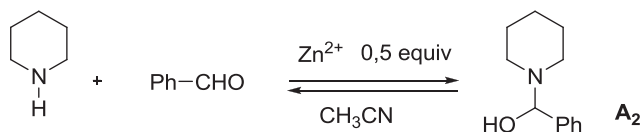


Fig. 9. Three-dimensional plot of the IR monitoring (3998–648 cm^{-1} area) corresponding to the experimental reaction conditions summarized in Fig. 7.

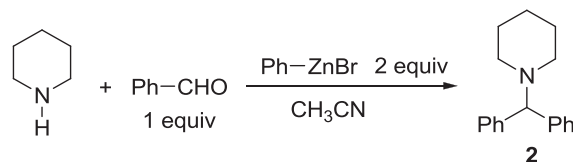


Scheme 8. Formation of hemiaminal **A**₂.

several minutes, and then preformed phenylzinc bromide [21] (2 equiv) was added dropwise to the solution (Scheme 9).

The monitoring, over time, of the absorbance for the bands mentioned above, is depicted in Fig. 11.

At $t = 3$ min, piperidine was added to acetonitrile followed by benzaldehyde at $t = 7.5$ min. Both reagents were allowed to react for 10 min. As before, no formation of iminium species **B**₂ could be noticed. Then, the first equiv of the organozinc was added dropwise to the reaction medium between $t = 17.5$ min and $t = 24$ min. As soon as the first drops of the organozinc solution had been added, the formation of hemiaminal **A**₂ could be identified by the noticeable appearance of the three previously identified representative bands



Scheme 9. Reaction scheme corresponding to the experimental protocol when using piperidine as a model secondary amine.

at 1084, 1006 and 3540 cm^{-1} , the band at 1274 cm^{-1} suffering from an overlap with other bands was not considered. The concentration of the hemiaminal (light-orange, orange and light-brown curves) jumped to a plateau until $t = 24$ min, corresponding to the complete addition of the first equiv of phenylzinc bromide. At $t = 24$ min, the second equiv of the organozinc was added quickly. As a result of this addition, the concentration of the hemiaminal rapidly decreased and the three-component coupling product **2** appeared over time, as confirmed independently by gas chromatography. These results support the preliminary formation of the intermediate hemiaminal **A**₂. This intermediate could then be involved in several reaction pathways. The first possibility would be to form the iminium ion by dehydroxylation. However, as mentioned above, no iminium ion could be noticed by FTIR. In addition, although it could be supposed that this species would be instantly trapped by the organozinc, chromatographic analyses of the reaction medium over time revealed that, while the organozinc reagent is continuously consumed upon addition to the reaction medium, no three-component coupling product is obtained before one equiv of the organozinc reagent had been added to the medium. Consequently, these elements are more compatible with a second alternate reaction pathway that would involve the consumption of the organozinc in a process implying another intermediate. A plausible mechanism for this consumption of the organozinc reagent could be envisaged through a sequence in which **A**₂ would be initially consumed by an acid–base reaction with the organozinc to form the alcoholate **A**₂' (Scheme 10).

Further observations support this hypothesis. Indeed, between 0 and 1 equivalent of added organozinc, benzaldehyde and piperidine are still present in the reaction mixture, implying that the hemiaminal is regenerated and maintain the monitored

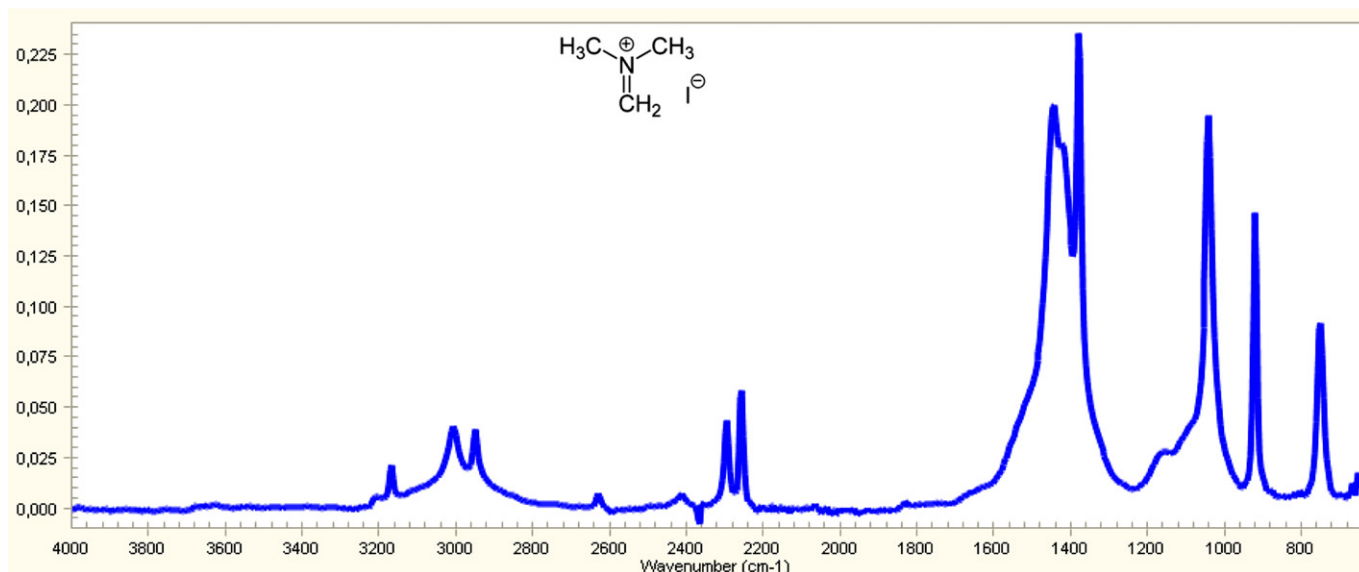


Fig. 10. IR spectrum of Eschenmoser's salt.

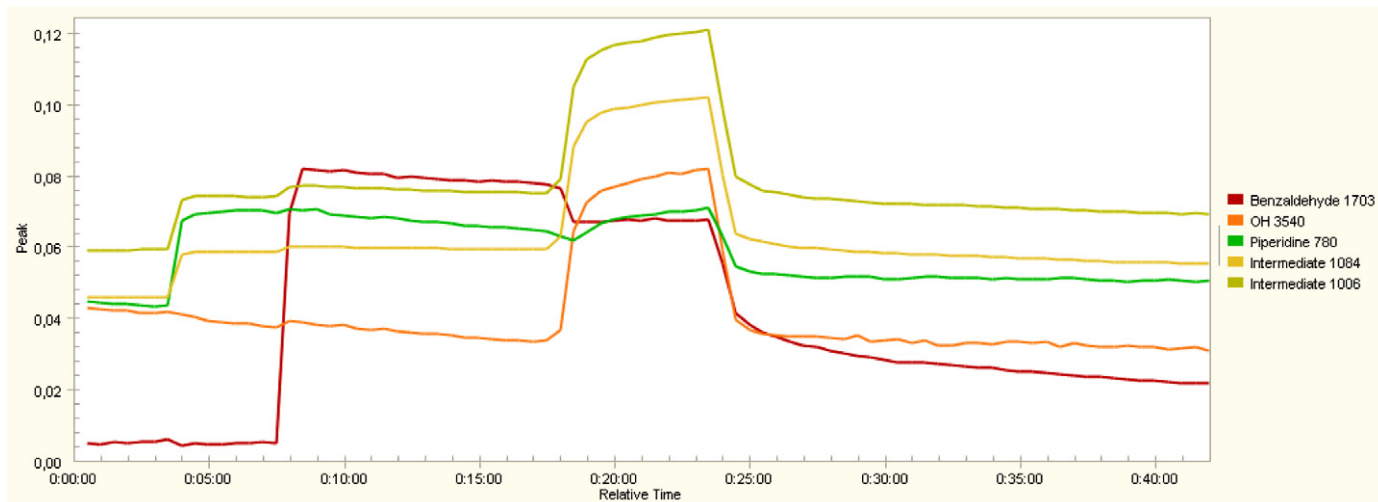
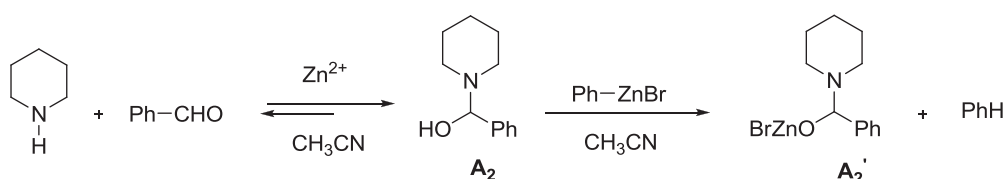


Fig. 11. Monitoring of the three-component reaction when using piperidine as a model secondary amine; subsequent addition to acetonitrile used as a solvent of: (1) piperidine (green curve (780 cm^{-1} band)), (2) benzaldehyde (red curve (1703 cm^{-1} band)), and (3) phenylzinc bromide; intermediate species appearance is monitored at 1006 cm^{-1} (light-brown curve), 1084 cm^{-1} (light-orange curve) and 3540 cm^{-1} (orange curve). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



Scheme 10. Successive formation of a hemiaminal and an alcoholate in the proposed mechanism.

concentration constant. Later, when benzaldehyde and piperidine are in insufficient concentration to counterbalance the consumption of the hemiaminal, the concentration of **A₂** quickly decreases as clearly evidenced in the 3500 cm^{-1} area of Fig. 12.

At this stage, one equiv of the zinc alcoholate **A₂'** might have been produced, hence additional organozinc introduced in the reaction medium should serve as a nucleophile in the following mechanistic steps. GC experiments confirm that the three-component product starts to appear at this stage. Thus, it seems plausible to envision a reaction of the organozinc with the alcoholate **A₂** via the 6-center transition state **TS** as postulated above (Scheme 11), via a zinc-assisted nucleophilic displacement of the oxygen by the organozinc and the final production of the coupling product **2**, along with zinc salts (path II in Scheme 11).

Consequently, this part of the study also provides useful information about the multicomponent reaction between a secondary amine, an aromatic aldehyde and an arylzinc reagent. Indeed, it can be argued that, contrary to what was observed with primary amines, the three-component reaction of secondary amines does not involve the formation of a C=N double bond by dehydroxylation and trapping by the organozinc reagent, but rather the formation of an alcoholate and a phenyl transfer at the stage of a 6-center transition state **TS**. This result is possibly consistent with a limited formation of the iminium intermediate due to a rapid acid–base reaction between the hemiaminal **A₂** and the organozinc agent.

Our results are finally to be compared with those described by Fan et al. for similar multicomponent coupling reactions realized in tetrahydrofuran. Indeed, it was supposed by these authors that, when using primary amines, the key intermediate of the reaction is

a hemiaminal, although not in acetonitrile, in which case an imine is produced as noticed in the present study. Consequently, the solvent might play a critical role in the reaction fate and mechanism. These findings are not surprising since we had already noticed that the three-component reaction of benzyl bromide with benzaldehyde and aniline is both quite more rapid (10–20 times faster) in CH_3CN than in THF, and that the reaction of arylzinc compounds is strongly inhibited in THF contrary to what is observed in CH_3CN [14].

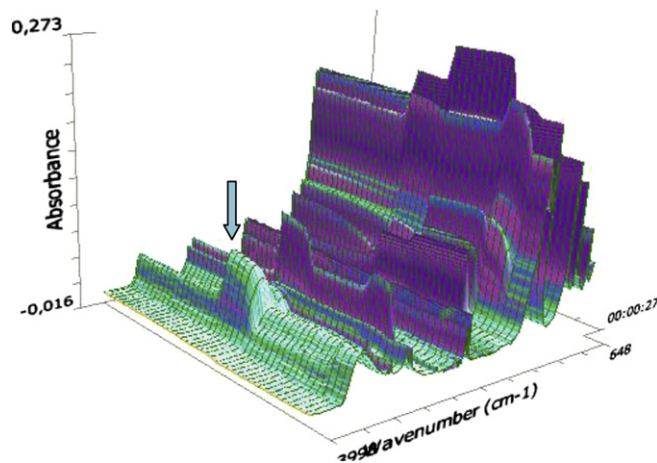
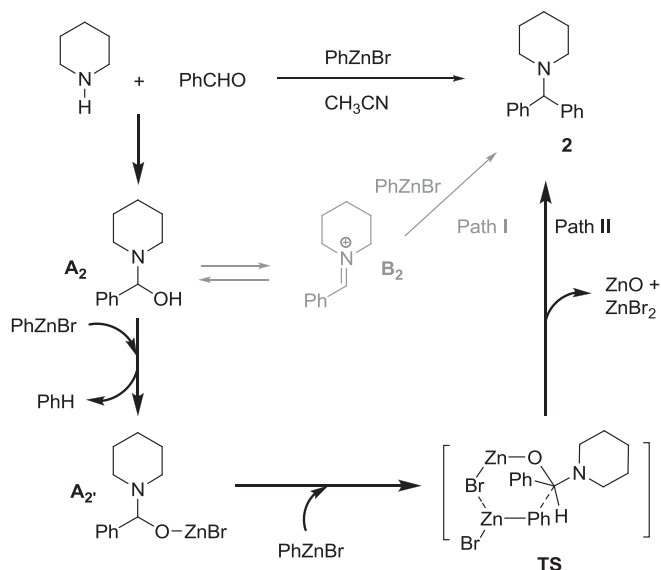


Fig. 12. Three-dimensional plot of the IR monitoring ($3998\text{--}648\text{ cm}^{-1}$ area) corresponding to the reaction conditions indicated in Fig. 11.



Scheme 11. Putative reaction pathways when secondary amines are used.

3. Conclusion

In conclusion, we have demonstrated by *in situ* spectroscopic experiments using a ReactIR instrument that the Mannich-like three-component reaction between amines, aromatic aldehydes and organozinc reagents might proceed according to two distinct pathways. While reactions involving primary amines proceed via *in situ*-formed imines, the reactions involving secondary amines rather feature a hemiaminal as the key intermediate. These results will be used for the further optimization of the multicomponent system and the design of stereoselective versions of the reaction.

4. Experimental section

Reactions were monitored by Fourier transformed infra-red (FTIR) *in situ* spectroscopy using a Mettler Toledo ReactIR iC10 fitted an articulated arm and a diamond probe. The ATR mode (attenuated total reflection) was used for the analyses, which were conducted in the 650–4000 cm^{-1} optical range. Spectral interferences were observed between 1950 and 2250 cm^{-1} , due to the intrinsic absorption of diamond in this area. The *iC IR* software was used for the monitoring of the whole system and infra-red spectra handling. In all cases, selection of the bands was performed manually and the optional extraction and monitoring of “components” by the *iC IR* software was not used (using this software

function, it is possible to automatically detect the formation or disappearance of compounds or intermediates whose simulated spectra are extracted from the overall signal). Reactions were concomitantly monitored by gas chromatography (GC) using a Varian 3400 chromatograph fitted with a capillary DB1 column ($l = 5 \text{ m}$, $\varnothing = 0.32 \text{ mm}$, $df = 0.4 \mu\text{m}$). Solvents and reagents were purchased from commercial suppliers and used without further purification. Before monitoring experiments, the IR spectra of solvent, starting and final compounds were recorded under neat conditions for comparison purpose. Monitoring experiments were then conducted in a 100 mL tricol under argon atmosphere. In a typical experiment, 30 mL acetonitrile were added to the flask, followed by the amine and the aldehyde (10 mmol of each). The organozinc compound was prepared independently on a 30 mmol scale, in 30 mL acetonitrile (see Refs. [13,14,20,21] for details). Then the filtered (syringe) solution was added dropwise to the medium under constant IR monitoring.

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